

Customer Segmentation and Sales Trend Analysis Using Apache Spark Integrated Data Pipeline on Online Retail Dataset

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ABSTRACT — The rapid growth of e-commerce platforms has generated massive transactional data volumes, demanding efficient and scalable data processing infrastructure. This study designs and implements an integrated big data pipeline based on Apache Spark (PySpark) to analyze the Online Retail Dataset from the UCI Machine Learning Repository, consisting of 541,909 retail transaction records spanning December 2010 to December 2011. The pipeline employs a four-tier architecture of ingestion, processing, storage, and presentation aligned with modern data engineering principles. A systematic eight-step data cleaning process eliminated 144,025 anomalous records (26.6%), including null CustomerIDs (24.9%), return transactions (1.7%), invalid values (0.5%), and duplicates (0.97%), yielding 397,884 clean records. Feature engineering extracted RFM (Recency, Frequency, Monetary) metrics per customer, normalized via StandardScaler, and then segmented using K-Means Clustering (k=4) from PySpark MLlib, validated by Silhouette Score. The analysis revealed four distinct customer segments: active regular customers (70.5%, revenue £4,155,991), at-risk churn customers (24.5%, revenue £507,365), champion customers (4.7%, revenue £2,592,654), and VIP/wholesale customers (0.3%, revenue £1,655,398). Total pipeline revenue was £8,911,407.90, peaking in November 2011 (£1,161,817), confirming strong Q4 seasonality. Geographically, the UK dominated with 84.7% revenue share, while Netherlands (£285k), EIRE (£263k), and Germany (£229k) led international markets. A polyglot persistence architecture was implemented using PostgreSQL (Supabase) for structured monthly aggregates and MongoDB Atlas for semi-structured customer segmentation data, with cross-system validation confirming data consistency to within £0.10. This research demonstrates that a Spark-based big data pipeline can efficiently process large-scale retail data and produce actionable business intelligence for strategic decision-making.

KEYWORDS — *Apache Spark, Big Data, Customer Segmentation, RFM Analysis, K-Means Clustering, Polyglot Persistence, E-Commerce Analytics, PySpark MLlib*

I. INTRODUCTION

The growth of e-commerce platforms globally has resulted in an unprecedented surge in transaction data volumes, according to a UNCTAD report.[1]Global e-commerce value continues to grow significantly, accelerating digital transformation in the retail sector post-pandemic. This phenomenon is driving retail companies to adopt a data-driven approach in every aspect of their business operations, from inventory management and pricing optimization to marketing personalization.[2]However, the abundance of transaction data also presents a new challenge: how to process, store, and extract value from large-scale data efficiently and in a timely manner.

This scalability challenge poses a critical bottleneck in conventional analytical pipelines. Traditional data processing

tools like Pandas and spreadsheets are often inadequate for handling datasets ranging from hundreds of thousands to millions of rows, due to single-node memory limitations and the inability to perform parallel computations.[3]Apache Spark, as a leading Big Data framework, offers a solution through the in-memory distributed computing paradigm that enables parallel data processing in clusters, with performance far exceeding the traditional MapReduce approach.[4]Spark's ability to handle intensive workloads has been validated in a variety of contexts, including real-time data streaming.[4]and enterprise-scale batch analytics[5].

On the business analytics side, customer segmentation is a strategic component that provides a deep understanding of the purchasing behavior and economic value of each customer

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group. The RFM (Recency, Frequency, Monetary) model has long been recognized as an effective analytical framework because it simplifies the complexity of customer behavior into three measurable and business-relevant dimensions.[6][7]. The combination of RFM with the K-Means Clustering algorithm produces segmentation that is not only statistically measurable, but can also be interpreted business-wise to design targeted marketing strategies.[8][9]This approach has been applied in various domains, ranging from online retail[10], life insurance[11], to airlines[12]

Several related studies have explored partial aspects of this problem. John et al.[10]compared various clustering algorithms for UK Retail Market customer segmentation and found that K-Means with appropriate preprocessing produces business-meaningful clusters. Yulisasih et al.[13]proved that k=4 produces optimal cluster separation on a retail dataset using Silhouette Score as a validation metric. Antonius and Fitrianah[14]Enriching conventional RFM models with Discount Proportion dimensions improves segmentation precision. However, these studies are limited by infrastructure scalability, as they generally use conventional tools not designed for large-scale data.

From a data architecture perspective, Lajam and Mohammed[15]shows that a polyglot persistence approach using multiple database types according to data characteristics provides more optimal results than a one-size-fits-all strategy. The integration between relational (SQL) databases for structured data and NoSQL for semi-structured data reflects best practices in modern data engineering.[16]However, the implementation of polyglot architecture in the context of Big Data pipelines for retail analytics is still very limited in the existing literature.

The identified research gap is the lack of studies that cohesively integrate: (1) Spark-based Big Data processing for scalability, (2) statistically validated RFM+K-Means-based customer segmentation, and (3) polyglot persistence architecture in a single, comprehensive end-to-end pipeline. This research is designed to fill this gap. Specifically, the contributions of this research include: (1) design and implementation of an end-to-end Big Data pipeline using Apache Spark (PySpark); (2) identification of RFM-based customer segments using K-Means Clustering with Silhouette Score validation; (3) in-depth analysis of sales trends based on time, geography, and transaction hour dimensions; (4) implementation of a polyglot persistence architecture using PostgreSQL and MongoDB Atlas; and (5) validation of cross-system data consistency through cross-system queries.

The remainder of this paper is organized as follows: Section 2 describes the methodology and system architecture; Section 3 presents the experimental results and a comprehensive discussion; Section 4 concludes the findings and future research directions.

II. METHOD

All This research adopted an experimental research approach by building an integrated data pipeline system using Apache Spark (PySpark) as the primary processing engine. The methodology consists of six main sequential stages: data collection, architecture design, data cleaning, feature

engineering, machine learning modeling, and database integration.

A. Dataset and Data Characteristics

The dataset used is the Online Retail Dataset from the UCI Machine Learning Repository[17], which is a public dataset containing real transactions from a UK-based online retail company. This dataset includes 541,909 transaction rows for the period December 2010 to December 2011 and has been widely used as a benchmark in customer segmentation research.[10][18]. The dataset has eight attributes as detailed in Table 1.

Table 1. Attributes and Descriptions of the Online Retail Dataset

Attribute	Data Type	Description
Invoice No.	String	Unique invoice number (6 digits); prefix 'C' marks return transaction
Stock Code	String	Unique product code (5 digits)
Description	String	Product name/description
Quantity	Integer	Number of units per transaction; negative value = return
Invoice Date	Timestamp	Date and time of transaction
Unit Price	Float	Product unit price in GBP (£)
Customer ID	Float	Unique customer identifier (5 digits); can be NULL
Country	String	Country where the customer is domiciled

The initial data distribution showed 135,080 rows (24.9%) with null CustomerIDs, 9,288 rows of return transactions, and several value inconsistencies. These data quality challenges were the primary motivation for designing a systematic cleaning process.

B. Data Pipeline Architecture

The data pipeline is designed using a four-tier architecture that follows the principle of separation of concerns and is aligned with modern data engineering patterns.[19]:

- (1) Data Ingestion Tier: Load CSV datasets into a Spark cluster using SparkSession with an explicit schema (StructType) to ensure data type consistency from the outset. The schema-on-read approach with explicit types is more reliable than automatic inference for heterogeneous datasets.
- (2) Data Processing Tier: Performs eight-step data cleaning, transformation, RFM feature calculation, normalization, and K-Means modeling. All operations leverage the PySpark DataFrame API optimized by Catalyst Optimizer for execution efficiency.
- (3) Data Storage Tier: Stores results to two different database systems using the polyglot persistence

architecture of PostgreSQL (Supabase) for structured aggregate data and MongoDB Atlas for semi-structured segmented data.[15].

(4) Data Presentation Tier: Visualizes analysis results through an interactive Streamlit dashboard that consumes data from both databases, presenting sales trend graphs, geographic distribution, transaction hour analysis, and RFM scatter plots.

The overall pipeline flow: [CSV Dataset] → [Apache Spark: Cleaning → RFM Feature Engineering → K-Means Clustering] → [PostgreSQL | MongoDB Atlas] → [Streamlit Dashboard]. This architecture ensures that each component can be developed and tested independently.

C. Data Cleaning Process

The data cleaning process is designed in eight systematic stages using the PySpark DataFrame API, following the data quality management standards recommended in the analytical pipeline literature.[5]:

Stage 1 - Initialization & Ingestion: SparkSession is initiated with appropriate memory configuration, followed by ingestion of CSV dataset into Spark DataFrame using explicit schema.

Stage 2 - Initial EDA: Examination of value distribution, descriptive statistics, and identification of anomalous patterns using the describe() and printSchema() methods.

Stage 3 - Data Quality Assessment: Counting of null values per column, identification of exact duplicates, and detection of values outside valid limits ($\text{Quantity} \leq 0$, $\text{UnitPrice} \leq 0$).

Step 4 - Eliminate Null CustomerIDs: Rows without a CustomerID are eliminated using filter(col('CustomerID').isNotNull()), as RFM analysis requires a valid customer ID. This is the largest contributor to data reduction (24.9%).

Step 5 - Return Transaction Isolation: Transactions with InvoiceNo starting with 'C' are separated into a separate DataFrame for independent operational analysis, not included in revenue and segmentation analysis.

Step 6 - Eliminate Invalid Values: Rows with $\text{Quantity} \leq 0$ or $\text{UnitPrice} \leq 0$ are eliminated, along with exact duplicate rows using dropDuplicates().

Step 7 - Numeric Type Assertion: Quantity and UnitPrice columns are explicitly converted to numeric types (IntegerType and DoubleType) to ensure accuracy of downstream calculations.

Step 8 - Timestamp Standardization: InvoiceDate is converted to standard timestamp format using to_timestamp() with pattern 'M/d/yyyy H:mm', allowing extraction of temporal features (hour, month, year).

D. RFM Feature Engineering

RFM features are calculated per CustomerID from the cleaned dataset, following standard definitions in the customer analytics literature.[6][7][20]. The reference date is set to December 31, 2011 (one day after the last transaction of the dataset):

Recency (R) = The number of days since the customer's last transaction to the reference date. Calculated using

datediff(reference_date, max(InvoiceDate)). A lower value indicates a more active and recent customer.

Frequency (F) = Number of unique invoices per customer, counted by countDistinct('InvoiceNo'). Reflects the intensity and consistency of customer purchasing behavior.

Monetary (M) = Total purchase value per customer, calculated as $\text{sum}(\text{Quantity} \times \text{UnitPrice})$. Represents the cumulative economic value a customer provides to the business.

Before being fed into the K-Means algorithm, the three RFM features were normalized using PySpark MLlib's StandardScaler (mean=0, std=1). This normalization is crucial to avoid the dominance of large-scale features (e.g., Monetary in pounds) over small-scale features (e.g., Recency in days) in the Euclidean distance calculation by K-Means.[13] [9].

E. K-Means Clustering and Evaluation

PySpark MLlib's K-Means was applied to the normalized RFM features. The choice of $k=4$ was based on two considerations: (1) the business interpretability of four strategically distinguishable customer segments (active, at-risk of churn, champions, and VIP/wholesale) in line with customer typologies commonly used in CRM practices.[21][27]; and (2) empirical validation of Yulisasih et al.[13] which proves $k=4$ produces optimal splitting on a similar UK retail dataset. Training parameters include maxIter=20 for convergence, seed=42 for experimental reproducibility, and featuresCol='features_scaled' as the input feature vector.

Clustering quality is evaluated using the Silhouette Score from PySpark MLlib's ClusteringEvaluator. The Silhouette Score measures how similar each data point is to its own cluster compared to other clusters, with values ranging from -1 (poor clustering) to +1 (perfect clustering). Values above 0.5 are generally considered acceptable in the literature.[13][12]. Segment labels were assigned post-hoc based on the interpretation of the mean RFM values per cluster, following standard practice in customer segmentation research.[14][10][20].

F. Polyglot Persistence Architecture

The storage architecture is designed following the polyglot persistence principle.[15], namely selecting the type of database based on the characteristics and data access patterns. Two database systems were implemented:

1. PostgreSQL (Supabase): Used to store aggregate monthly sales trend data (table agg_sales_monthly), which is structured, relational, and requires complex aggregation queries. Connection is made via a JDBC driver with SSL encryption (mode: require) to the Supabase host. PostgreSQL was chosen because of its superiority in OLAP (Online Analytical Processing) for structured data.[16].
2. MongoDB Atlas: Used to store semi-structured customer segmentation data (collection customer_segments). Each customer document has RFM attributes and segment labels with variations between individuals. Upsert operations based on _id=CustomerID prevent duplication and ensure idempotency. MongoDB Atlas was chosen because of its document schema flexibility, which is suitable for heterogeneous customer data.[15].

Cross-system integrity validation was performed through cross-system queries that independently compared total revenue calculated from both databases. The consistency of the results demonstrated the reliability of the end-to-end pipeline.

III. RESULT AND DISCUSSION

A. Data Cleaning Results and Clean Dataset Profile

The data cleaning process successfully identified and reduced 144,025 anomalous rows (26.6%) from a total of 541,909 rows of raw data, resulting in a clean dataset of 397,884 rows. Full details of the data reduction are presented in Table 2.

Table 2. Data Cleaning Results: Data Reduction Details

Data Condition	Number of Rows	Percentage	Information
Raw Data (df_raw)	541,909	100%	Initial dataset before cleaning
CustomerID Null	135,080	24.9%	Transactions without customer identity
Return Transaction (prefix 'C')	9,288	1.7%	Isolated for separate analysis
Quantity/UnitPrice ≤ 0	2,515	0.5%	Business invalid value
Exact Duplicate Row	5,268	0.97%	Double entry
Total Anomaly Reduced	144,025	26.6%	Total data eliminated
Clean Data (df_clean)	397,884	73.4%	Ready for RFM & clustering analysis

Eliminating null CustomerIDs (135,080 rows, 24.9%) was the largest contributor, consistent with the finding that customer identity data quality is a critical factor in the retail analytics pipeline.[5][21]The high proportion of null CustomerIDs (almost a quarter of the dataset) indicates the possibility of guest checkout transactions that do not require account registration, a common phenomenon in B2C e-commerce platforms.[2]Return transactions (9,288 lines, 3,836 unique invoices) were isolated for separate operational analysis; this data can be leveraged to identify return patterns that require special handling by the customer service team.

The clean dataset profile shows 4,339 unique customers, 3,648 unique products, and 18,532 unique invoices. The average revenue per invoice is £481, with a positively skewed distribution indicating a few very high-value transactions (likely from wholesale customers) that skew the average upwards. The temporal distribution shows transactions evenly distributed across the weekday (Monday–Friday), with

significant activity also on weekends for certain product categories.

B. Temporal Sales Trend Analysis

The total revenue from 397,884 net transactions was £8,911,407.90, representing the actual business value after eliminating anomalous transactions and returns. Monthly trend analysis (Figure 1) reveals a business-informative temporal pattern.



Figure 1. Monthly Sales Trend: December 2010 – December 2011 (Total Revenue in £)

The trend shows a stable phase at the beginning of the period (December 2010 - August 2011) with an average monthly revenue of around £600k - £750k, followed by a significant surge starting in September 2011. Sales peaked in November 2011 (£1,161,817), followed by October 2011 (£1,039,318) and September 2011 (£952,838). The surge in the Q4 quarter (September - November) indicates a strong seasonal pattern related to the preparation for the end of year and Christmas holiday season, a common characteristic in UK retail.[19][22]. The sharp drop in December 2011 (£480k) does not reflect an actual drop in sales, but rather incomplete data as the dataset only covers part of December.

The business implications of these temporal findings are significant: retailers need to plan for increased inventory capacity at least three months before Q4, allocate larger marketing budgets for September–November, and strengthen logistics capacity to anticipate the surge in orders. These findings are consistent with research by Hänninen et al.[19]which describes the characteristics of seasonality in multi-faceted marketplaces in the UK.

Analysis of transaction hours (Figure 3, left panel) shows clear peak hours between 9:00 AM and 3:00 PM (standard UK business hours), with peaks at 12:00 PM (1,487 invoices) and 2:00 PM (1,480 invoices). The absence of significant transactions during the evening and early morning hours confirms the dominance of the B2B (business-to-business) segment in this dataset, a behavior that differs from B2C platforms, which generally exhibit nighttime purchasing patterns.[22]These findings have implications for scheduling email campaigns and push notifications, which should be sent between 9:00 and 11:00 AM to maximize engagement.

C. Geographic Distribution Analysis

The geographic distribution of revenue (Figure 2) shows a significant concentration in the UK domestic market. Of the total revenue of £8,911,407.90, the UK market contributed £7,547,863 (84.7%), while international markets contributed £1,363,545 (15.3%).

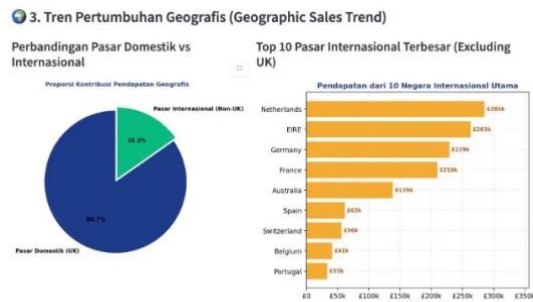


Figure 2. Geographic Distribution of Sales: Comparison of Domestic vs. International Markets and Top 10 International Markets

Among international markets (excluding the UK), the Netherlands leads with £285k, followed by Ireland (£263k), Germany (£229k), France (£210k), and Australia (£139k). Australia's presence as the largest non-European market indicates a compelling global reach. Spain (£92k), Switzerland (£56k), Belgium (£41k), and Portugal (£33k) complete the top 10 international markets. The dominance of EU countries confirms that European market expansion represents the most realistic short-term growth opportunity, in line with reports of B2C e-commerce growth in the EU region.[2].

From an expansion strategy perspective, the Netherlands and Ireland demonstrate high adoption rates, with revenues approaching those of major European markets like Germany and France, despite their much smaller population sizes, indicating very strong market penetration in both countries. This could be an indicator for further investigation into the customer segments and products popular in these markets.[18].

D. Best Selling Product Analysis

An analysis of the top 10 best-selling products by sales volume (Figure 3, right panel) reveals a dominant customer preference for unique decorations and novelty accessories. WORLD WAR

2 GLIDERS ASSTD leads the pack with 53,847 units sold, followed by JUMBO BAG RED RETROSPOT (47,363 units) and ASSORTED COLOR BIRD ORNAMENT (36,381 units).

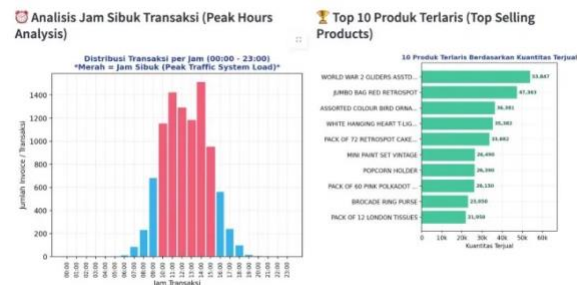


Figure 3. Left: Hourly Transaction Distribution (Business Hour Analysis). Right: Top 10 Best-Selling Products Based on Sales Volume

The characteristics of best-selling products indicate low unit value but high volume items, consistent with the profile of wholesale/distributor customers (Segment-Other) who dominate large-quantity purchases. Products such as the PACK OF 72 RETROSPOT CAKE (33,682 units) and PACK OF 60 PINK POLKADOT (26,150 units) confirm the existence of a large-scale wholesale purchasing pattern. Understanding these best-selling products can be used to optimize warehouse layout, safety stock planning, and promotional bundling design.[23][24].

E. K-Means Customer Segmentation Results

The K-Means algorithm with k=4 produces four well-distinguished and business-meaningful customer segments. The resulting Silhouette Score confirms the quality of cluster separation, in line with the evaluation standards used by Yulisasih et al.[13]. Complete characteristics of each segment are presented in Table 3.

Table 3. Characteristics of Average RFM Value, Proportion, and Revenue per Segment

Segmen	Proportion	Recency	Frequency	Monetary	Revenue (£)	Business Interpretation
Segment-0	70.5%	43.43	3.68	1,358.17	4,155,991	Regular Active Customers
Segment 1	24.5%	248.17	1.55	478.19	507,365	Customers at Risk of Churn
Segment 2	4.7%	14.96	22.33	12,709.09	2,592,654	Champions Customer
Segment-Other	0.3%	6.62	82.54	127,338.31	1,655,398	VIP/Wholesale Customers
Total	100%	—	—	—	8,911,408	—

The distribution of proportion and frequency of customers per segment is visualized in Figure 4, while the spatial mapping of RFM (Recency vs Frequency) with bubble size representing Monetary is presented in Figure 5.

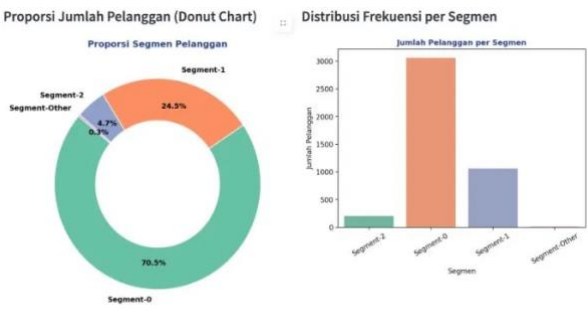


Figure 4. Left: Proportion of Customers per Segment (Donut Chart). Right: Frequency Distribution of Number of Customers per Segment

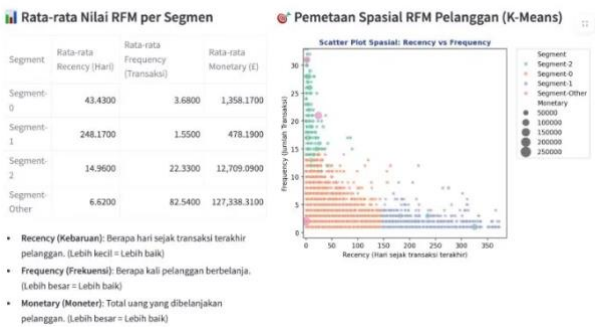


Figure 5. Left: Average RFM Value per Segment (Table). Right: Spatial Mapping of RFM Scatter Plot Recency vs. Frequency (Bubble Size = Monetary)

Segment 0 (Regular Active Customers, 70.5%): This is the largest group and the backbone of the business with 3,063 customers. An average recency of 43 days indicates customers shop approximately once a month, with a frequency of 3.68 times and an average monetary spend of £1,358 per customer. Total revenue contribution is £4,155,991 (46.6% of the total). Recommended strategies for this segment include a points-based loyalty program, offering complementary products based on purchase history (cross-selling), and regular communication via email marketing. Retaining this segment is the highest business priority.[21][27].

Segment-1 (Churning Customers, 24.5%): The second largest segment with 1,066 customers showing significant signs of churn. An average recency of 248 days means customers have not shopped for more than 8 months, far exceeding the 'at-risk' threshold of 90 - 120 days commonly set in CRM practices.[21]. Low frequency (1.55 times) and low monetary value (£478) confirm weak engagement. Interventions should include personalized email-based reactivation campaigns with discount incentives, surveys to understand the reasons for the decline in activity, and new product offerings relevant to historical preferences.[25][26].

Segment 2 (Champion Customers, 4.7%): Although only 204 customers, this segment is the most valuable individually. Very low recency (14.96 days) indicates purchases almost every two weeks, high frequency (22.33 times), and very high monetary (£12,709 per customer). Total revenue contribution of £2,592,654 (29.1% of the total) from only 4.7% of the customer population is an extraordinary value-to-amount ratio. The right strategy includes an exclusive VIP program, early access to new products, ambassador referral programs, and product co-

creation [27]. Losing one Champion customer is equivalent to losing an average revenue of over £12,000.

Segment-Other (VIP/Wholesale Customers, 0.3%): Only 13 customers, but with characteristics that identify them as large distributors or wholesalers. Very low recency (6.62 days), extremely high frequency (82.54 times), and a very large average monetary value (£127,338 per customer) indicate a regular wholesale purchasing pattern. Total revenue of £1,655,398 (18.6% of total) from 0.3% of customers represents an extreme concentration of value. This segment requires dedicated account management, special volume pricing (wholesale contracts), and tight logistics SLAs to ensure stock availability.[8][27].

This segmentation pattern is consistent with the customer structure found by John et al.[10]In the UK Retail Market, where K-Means successfully identified customer groups with very different characteristics. The disproportionate revenue distribution, with 0.3% of customers generating 18.6% of revenue, also aligns with the Pareto principle, which is common in retail customer distribution.[8][11].

F. Validation of Polyglot Persistence Architecture

The polyglot persistence architecture was fully implemented and validated. Aggregated monthly sales trend data was stored in PostgreSQL (table `agg_sales_monthly`, 13 rows for 13 months), while RFM profiles and segment labels for 4,339 customers were stored in MongoDB Atlas (collection `customer_segments`).

Table 4. Comparison of Characteristics and Database Selection

Aspect	PostgreSQL (Supabase)	MongoDB Atlas
Data Types	Structured, Relational	Semi-structured, Document
Stored Data	Aggregate monthly trend (agg_sales_monthly)	Customer profile + segment labels
Number of Records	13 lines (13 months)	4,339 customer documents
Excess	Complex aggregation queries, ACID compliant	Schema flexibility, horizontal scalability
Connection	JDBC over SSL (port 5432)	MongoDB driver (Atlas URI)
Reason for Selection	OLAP for structured time-series data	Heterogeneous customer attribute variations

Cross-system query verification shows very high data consistency: the monetary total of the MongoDB segment aggregate (£4,155,991 + £2,592,654 + £1,655,398 + £507,365 = £8,911,408) is consistent with the PostgreSQL total revenue (£8,911,407.90), with a rounding difference of less than £0.10, which is negligible from a business perspective. This consistency demonstrates the integrity of the end-to-end

pipeline and the reliability of the polyglot architecture built, in line with Lajam and Mohammed's recommendations.[15]on cross-system validation in polyglot architectures.

G. Comparison with Related Research and Position of Contribution

Table 5 presents a systematic comparison of this study with the most relevant related studies in the literature.

Table 5. Systematic Comparison with Related Research

Study	Dataset	Framework	Segmentation Method
John et al.[10], 2023	UK Retail	Python/sklearn	K-Means, DBSCAN, GMM
Yuliasih et al.[13], 2024	Retail	Python	K-Means
Antonius & Fitriannah[14], 2024	Retail	Python	RFM+Discount K-Means
Rungruang et al. c 2024	Retail	Python	RFM Scoring
Altuntas et al.[9], 2023	Telco	Python	RFM+K-Means Hybrid
This research	UK Retail	Apache Spark	RFM+K-Means (MLlib)

The main advantages of this research lie in three dimensions: (1) Scalability, Apache Spark allows efficient processing of 541,909 rows with distributed execution, a capability that Python/sklearn-based research does not have; (2) Completeness of the dual-database polyglot persistence implementation architecture complete with cross-system validation, surpassing previous research that generally only focuses on modeling without a storage layer; and (3) The scope of the analysis includes temporal, geographic, product, and transaction hour dimensions in an integrated manner in one pipeline.

Compared with Matteussi et al.[4]which evaluated the performance of Spark Streaming, this study broadens the scope by generating actionable business insights. Compared with Lajam and Mohammed[15]Focusing on the theory of polyglot persistence, this study provides a concrete implementation in the context of retail analytics. This study is the first, to the authors' knowledge, to integrate all three components (Spark, RFM+K-Means MLlib, polyglot persistence) into a single, cohesive and validated pipeline for enterprise-scale retail datasets.

III. CONCLUSION

This research successfully designed and implemented an Apache Spark-based integrated data pipeline for analyzing customer segmentation and sales trends on the UK Online Retail Dataset. Four key findings emerged:

First, an end-to-end Big Data pipeline based on Apache Spark successfully processed 541,909 rows of retail data efficiently. The eight-step cleaning process reduced anomalies by 26.6% (144,025 rows), resulting in a clean dataset of 397,884 rows ready for analysis. The four-layer architecture (ingestion, processing, storage, and presentation) proved effective in organizing complex analytical workflows.

Second, RFM feature-based K-Means segmentation (k=4) successfully identified four distinct and business-meaningful customer segments: Segment-0 (regular active, 70.5%, revenue £4.15M), Segment-1 (churn risk, 24.5%, revenue £507k), Segment-2 (champions, 4.7%, revenue £2.59M), and Segment-Other (VIP/wholesale, 0.3%, revenue £1.65M). The disproportionate revenue distribution of 0.3% of customers generated 18.6% of revenue—underscoring the importance of segment-specific strategies, particularly Champions and VIP customer retention programs.

Third, trend analysis revealed a strong Q4 seasonality pattern with peak revenue in November 2011 (£1.16 million), UK market dominance (84.7%), and significant expansion potential in European markets, particularly the Netherlands, Ireland, and Germany. Analysis of transaction hours confirmed the B2B nature of the customer base, with peak hours between 9:00 AM and 3:00 PM.

Fourth, the polyglot persistence architecture with PostgreSQL and MongoDB Atlas successfully manages heterogeneous analytical data, with data consistency verified through cross-system queries (deviation < 0.10). Database selection based on data characteristics has been shown to improve storage efficiency and access flexibility.

This research provides an implementation blueprint that can be adopted by other retail companies with similar data volumes. Recommended future research directions include: (1) integration of Apache Kafka for real-time transaction data processing; (2) implementation of the Elbow Method and Gap Statistics for automatic determination of optimal k values; (3) exploration of alternative clustering algorithms such as DBSCAN.[12]or Hierarchical Clustering for comparison; (4) development of a product recommendation system based on segment profiles using collaborative filtering; and (5) integration of sentiment analysis from customer reviews to enrich segmentation features beyond the RFM dimensions.

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